Akoizo Original Practice - Form 5

Division C | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which is quantitative evidence?

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Means increased from 12 to 20 units

B. It looks nice

C. The hypothesis is obviously true

D. The team worked hard

Answer: _____

2. (2 points) Randomizing order mainly helps address…

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. Every possible confounder automatically

B. Time-linked systematic differences

C. All measurement uncertainty

D. The need for units

Answer: _____

3. (2 points) 2.0 x 3.00 cm^2 is reported as…

A. 6.00000 cm^2

B. 6 cm

C. 5.0 cm^2

D. 6.0 cm^2

Answer: _____

4. (2 points) High R^2 proves causation?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

A. No

B. Yes

C. Only with a colored line

D. Only when n = 3

Answer: _____

5. (4 points) Name the example independent and dependent variables.

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

Answer: _____
_____
_____

6. (4 points) Why do more repeats not remove a fixed offset?

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

Answer: _____
_____
_____

7. (4 points) Why separate raw and summary data?

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

Answer: _____
_____
_____

8. (4 points) How can roles create contradictions?

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

Answer: _____
_____
_____

9. (4 points) Why show with/without-outlier results?

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

Answer: _____
_____
_____

10. (4 points) Why does replication not solve drift alone?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

Answer: _____
_____
_____

11. (4 points) Why check the applicable checklist?

Unit: Division C comparison standards, significant figures, and abstracts.
Worked-case data: Report area from measured lengths 2.0 cm and 3.00 cm, then report their sum.

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: For response values 8, 10, 12, calculate mean, range, and sample SD.

Answer: _____
_____
_____

13. (2 points) Define “Independent variable” in the context of Experimental Design.

Answer: _____
_____
_____

14. (2 points) Define “Random error” in the context of Experimental Design.

Answer: _____
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15. (2 points) Define “Raw data” in the context of Experimental Design.

Answer: _____
_____
_____

16. (2 points) Define “Coherence” in the context of Experimental Design.

Answer: _____
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17. (2 points) Define “Reasoning” in the context of Experimental Design.

Answer: _____
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_____

18. (2 points) Define “Confounding” in the context of Experimental Design.

Answer: _____
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